Adaptive Portfolio Intelligence: Explainable Reinforcement Learning for Regime-Aware Portfolio Allocation and Risk Management

Parv Patel (142301041) • Mandar Patil (112401021)

I. Abstract

Financial markets are dynamic and uncertain, which makes portfolio allocation a challenging decision-making problem. Traditional investment strategies often rely on fixed allocation rules or assumptions that may not adapt well to changing market conditions, transaction costs, and variations in risk. This project proposes *Adaptive Portfolio Intelligence*, a reinforcement learning-based framework for dynamic portfolio allocation and risk management.

The project will formulate portfolio management as a sequential decision-making problem. An agent will use information derived from historical market data to select portfolio allocations and receive feedback based on investment performance and risk. The study will investigate market conditions such as rising, falling, and high-volatility periods, and examine whether an adaptive policy can respond to these changes. Proximal Policy Optimisation (PPO), an actor–critic reinforcement learning method, will be considered as a candidate algorithm, alongside suitable baseline methods [1, 2].

The framework will be evaluated against conventional portfolio allocation benchmarks using measures such as cumulative returns, Sharpe ratio, maximum drawdown, and portfolio turnover. The evaluation will include out-of-sample testing and, where feasible, walk-forward validation and stress testing across different market conditions. Transaction costs will be considered to make the evaluation more realistic. Explainability techniques will also be explored to understand the factors associated with the agent's allocation decisions and risk-taking behaviour.

The expected outcome is a reproducible framework for studying the usefulness and limitations of reinforcement learning in adaptive portfolio management. The project will assess not only investment performance, but also risk control, adaptability, and the interpretability of the decisions made by the learned policy. The findings will be based on experimental evaluation; no improvement in performance is assumed in advance.

II. References

[1] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal Policy Optimization Algorithms," *arXiv preprint arXiv:1707.06347*, 2017.

[2] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. Cambridge, MA, USA: MIT Press, 2018.

[3] J. Moody and M. Saffell, "Learning to trade via direct reinforcement," *IEEE Transactions on Neural Networks*, vol. 12, no. 4, pp. 875–889, 2001.

III. AI Tools Disclosure

ChatGPT (OpenAI) was used to assist with language refinement and preparation of the LaTeX source. The project design, implementation, experiments, and interpretation of results will be carried out and verified by the student team.